BTEC Level 3 Applied Science · Unit 1 Physics · Answer sheet

Wave Properties — Answers

Mark scheme for the companion worksheet. Accept equivalent correct wording. These are original JDScience items, not official Pearson questions.

A–E

1. Transverse: oscillation perpendicular to energy transfer. Longitudinal: parallel.
2. Light / water / string; sound / P-waves / slinky compressions.
3. Resultant displacement is the sum of individual displacements.
4. Compression: particles closer / higher pressure. Rarefaction: more spaced / lower pressure.
5. They reach maxima and minima together / separated by $n\lambda$.
6. Light is transverse so oscillations have a direction that can be filtered; sound is longitudinal.
7. Ultrasound longitudinal; microwaves transverse; stadium sound longitudinal.
8. Yes — separation is 3λ, a whole number of wavelengths.
9. See definitions plus one valid example of each. [4]
10. Resultant displacement = sum of individual displacements. [2]
11. Polarisation needs a transverse direction; sound is longitudinal. [2]
12. The sine curve is a graph of pressure or displacement. The air particles still oscillate parallel to the direction of travel.